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GEOLOGY, GEOCHEMISTRY AND PALEOMAGNETISM OF THE CHENEAUX METAGABBRO (HELIKIAN) OF SOUTHERN QUEBEC AND EASTERN ONTARIO

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ABSTRACT

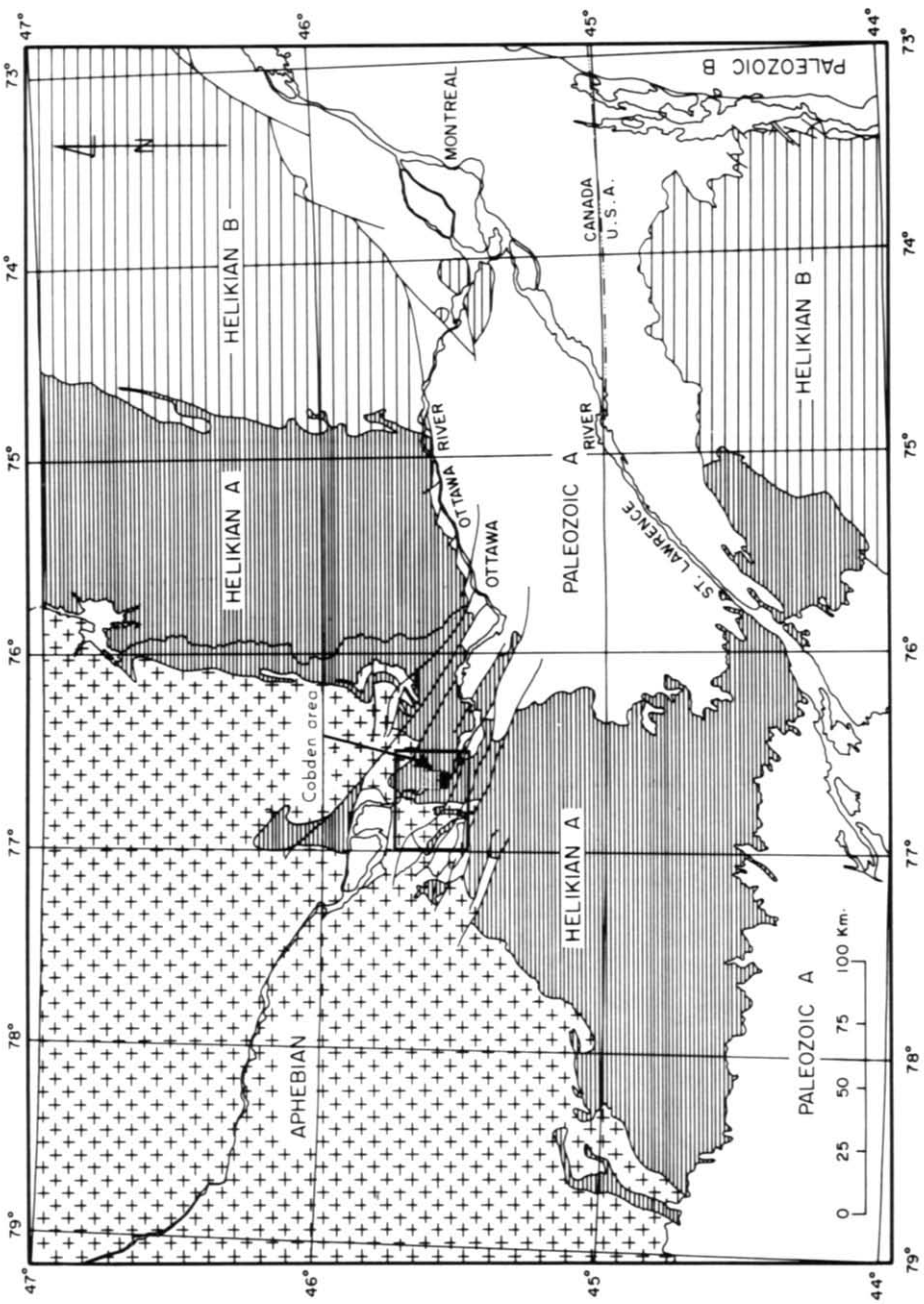
Seguin, M.K. and Brun, J., 1984. Geology, geochemistry and paleomagnetism of the Cheneaux metagabbro (Helikian) of southern Quebec and eastern Ontario. Precambrian Res., 26: 307—331.

The Cheneaux metagabbro (Helikian) is a slightly to moderately metamorphosed intrusive body in the southern Grenville province. The paleomagnetic study undertaken on it yielded 3 main components of magnetization. The first component (W+, D = 249°, I = +45°) with a corresponding paleopole of 44°E, 6°S may represent the time of emplacement or an early phase of mechanical deformation. The second component (W–, D = 258°, I = –34°) with a corresponding paleopole of 8°E, 21°N represents a main metasomatic event related to the initial phase of the Grenville orogeny. The third component (SE, D = 135°, I = 30°) with a corresponding paleopole of 32°W, 16°S is related to the final phase of the Grenville orogeny. The estimated age of acquisition of magnetisation for these 3 components are –1300, –1075 and –950 Ma, respectively. The significance of the new set of data in relation to the established Grenville apparent polar wandering curve is discussed. Finally a fourth component (NE, D = 37°, I = +41°) with a corresponding paleopole of 37°E, 53°N represents a much younger event; its pole may be located on the late Hadrynian track (–750 Ma).

INTRODUCTION

The Cheneaux metagabbro straddles the Quebec—Ontario boundary in the vicinity of Portage-du-Fort, Quebec (Fig. 1) and occurs near the contact between Aphebian rocks to the west and Helikian rocks to the east (Satterly, 1945; Lumbers, 1976a, 1977; Douglas, 1977). Intruding all Helikian metasedimentary units, it is considered as Helikian in age. The geology indicates that, after emplacement, it was involved in at least one episode of medium grade dynamic and thermal metamorphism. As a result, parts of it are highly altered while others remain almost unaltered. Because of this it was

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N
HELIKIAN B
HELIKIAN A
APHEBIAN
Cobden area
OTTAWA RIVER
OTTAWA
MONTREAL
CANADA
U.S.A.
PALEOZOIC A
ST LAWRENCE RIVER
HELIKIAN A
PALEOZOIC A
HELIKIAN B
PALEOZOIC B
0 25 50 75 100 Km.
47°
46°
45°
44°
73°
74°
75°
76°
77°
78°
79°

chosen as the subject for a paleomagnetic study. Nine sites were sampled and analyzed geologically, geochemically and paleomagnetically. The present paper gives the results of these analyses and discusses their significance in relation to the Grenville apparent polar wandering curve.

GEOLOGY

Regional geology (Cobden area)

The Cobden area (Fig. 2) forms part of the southern Grenville province. The Proterozoic rocks are covered by small isolated outliers of Orodovician rocks and widespread Pleistocene deposits. Four tectonostratigraphic units separated by 3 major disconformities are distinguished. These are: (1) the Aphebian (Early Proterozoic) which extends west of a line stretching from

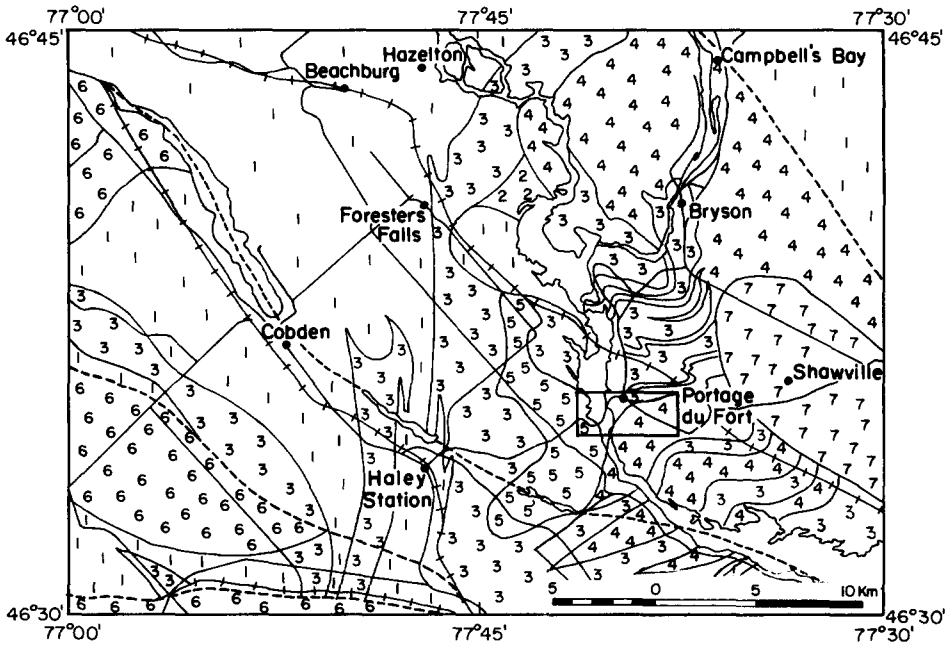


Fig. 2. Generalized geologic map of Cobden area; (1) Aphebian, granite gneiss, migmatite, granite (2) Helikian, metavolcanics (3) Helikian, carbonate metasedimentary Portage-du-Fort Group (4) Helikian, silicate metasedimentary Bryson Group (5) Helikian, mafic metaigneous Group (6) Ordovician, consolidated rocks (7) Pleistocene, unconsolidated sedimentary deposits. Compiled from maps of Osborne (1944), Satterly (1945), Katz (1976) and Brun (1983, in press).

Fig. 1. Simplified geologic map of eastern Ontario and southern Quebec: Aphebian granulite terrane, Helikian (A) amphibolite terrane, Helikian (B) granulite terrane, Paleozoic (A) unfolded terrane, Paleozoic (B) folded terrane. (Modified after Douglas, 1977.)

Hazelton (Westmeath township, Ontario) northward to Northcote (Admaston Township, Ontario) southward as illustrated by Satterly (1945) and Douglas (1977). It consists of folded metasedimentary and metaigneous rocks which exhibit extremely complex structural patterns of the Hudsonian (≈ 1800 Ma) orogeny and possibly Elsonian (≈ 1400 Ma) event (Krogh and Davis, 1969; Baer, 1976; Douglas, 1977). These rocks, deformed mostly into southeast-trending structures (Douglas, 1977), are characterized by at least 5 episodes of deformation and several episodes of migmatization (Appleyard, 1974). Mineral assemblages indicate that the metamorphism culminated at the pressure and temperature of the pyroxene granulite facies (Katz, 1976); (2) the Helikian (Middle Proterozoic) which occurs mainly east of the line stretching from Hazelton to Northcote and rests unconformably on Aphebian rocks (Douglas, 1977). It consists of folded metasedimentary and metaigneous rocks which exhibit complex structural patterns of the Elzevirian (ca. 1225—1100 Ma) and Ottawan (1050—1000 Ma) orogenies (Douglas, 1977; Moore and Thompson, 1980). These rocks, deformed mostly into northeast-trending structures (Krogh and Hurley, 1968), are characterized by at least 3 episodes of deformation and 2 episodes of migmatization (Appleyard, 1974). Mineral assemblages indicate that the metamorphism culminated at the pressure and temperature of the middle to upper almandine amphibolite facies (Lumbers, 1978); (3) the Ordovician which occurs mainly in the western part of the area, and rests unconformably on Helikian and Aphebian rocks. It consists of consolidated and unfolded rocks: sandstones and dolostones (Satterly, 1945; Katz, 1976); and (4) the Pleistocene which is ubiquitous and rests unconformably on Ordovician, Helikian and Aphebian rocks. It consists of unconsolidated and unfolded sedimentary deposits: clay, silt, sand, gravel and boulders (Satterly, 1945; Katz, 1976).

Local geology (Portage-du-Fort area)

The Portage-du-Fort area (Fig. 3) comprises all geological units mentioned previously except the Aphebian which is covered by the Helikian and outcrops only farther west in Ross township, Ontario. The Helikian stratigraphy includes, from bottom to top, the 5 following units:

(1) the metavolcanic group, which consists of various amphibolites observed west of the Ottawa river near Forester Falls (Ross township, Ontario) (Satterly, 1945) but missing east of the river (absent in Fig. 3). These rocks could be correlative with basic metavolcanics of the Hermon Group defined by Lumbers (1967a,b). In the Clarendon—Dalhousie area, the metavolcanic rocks were considered to be the oldest by Smith (1958). Moreover, in Tudor township, Ontario, the lower part of these metavolcanics has been dated (zircons by the U—Pb method) at 1310 ± 15 Ma (Silver and Lumbers, 1966);



Fig. 3. Detailed geologic map of Portage-du-Fort area, eastern Ontario and southern Quebec (Brun, 1983, in press). Gabbro sampling sites correspond to broken rectangle.

(2) the carbonate metasedimentary Portage-du-Fort Group, which consists of calcitic marbles, dolomitic marbles and calc-silicate marbles. These rocks occur west of the Ottawa river in Ross and Horton townships, Ontario, and east in Litchfield and Clarendon townships, Quebec, as indicated by Satterly (1945), Katz (1976) and Brun (1983, in press). These rocks could be correlative with marbles of the Mayo Group defined by Lumbers (1967a, b). In the Bancroft—Madoc area, the Mayo Group is considered to overlie the Hermon Group (Lumbers, 1967a,b);

(3) the silicate metasedimentary Bryson Group, which consists of quartzite, biotite gneiss, biotite and hornblende gneiss and amphibolite. These rocks occur west of the Ottawa river in Ross and Horton townships, Ontario, and east in Litchfield and Clarendon townships, Quebec, as indicated by Satterly (1945), Katz (1976) and Brun (1983, in press). These rocks could be correlative with gneiss of the Apsley unit defined by Simony (1960), Hewitt (1961) and Shaw (1962). In the Chandos—Methuen area, the Apsley gneiss overlies the Dungannon marble but is considered to be overlain by the Lasswade marble (Hewitt, 1961; Shaw, 1962).

(4) the mafic metaigneous group, which consists of metagabbros and metadolerites (including the Cheneaux metagabbro). These rocks occur west of the Ottawa river in Ross township, Ontario, and east in Litchfield township, Quebec, as indicated by Satterly (1945), Katz (1976) and Brun (1983, in press). These rocks could be correlative with the Umfraville, Grimsthorpe, Boulter, Thanet, Tudor, Dalhousie and Raglan metagabbro bodies in Ontario. Most of them intrude the metavolcanics, carbonate metasedimentary and silicate metasedimentary groups (Hewitt, 1956; Lumbers, 1967a, 1969; Palmer and Carmichael, 1973); and

(5) the felsic metaigneous group, which consists of metagranites, metaalaskites and pegmatites. These rocks occur east of the Ottawa river in Litchfield township, Quebec, as indicated by Brun (1983, in press). These rocks could be correlative with the felsic metamagmatic bodies of eastern Ontario. According to Hewitt (1956) all felsic bodies postdated the Raglan metagabbro in Raglan township, Ontario, but the Cordova Mines gabbro is considered to postdate pre-metamorphic bodies.

According to Katz (1976), the metamorphism of the Portage-du-Fort area is situated in the upper amphibolite (sillimanite—almandine—orthoclase) subfacies of the intermediate pressure facies series of the Barrovian type (Hietanen, 1967).

Pluton geology (Cheneaux metagabbro)

The pluton occupies parts of Horton and Ross townships, Ontario as well as Litchfield township, Quebec, but the largest part lies in Ontario just west of the Ottawa river (Fig. 2). Its length is ~ 11 km and its maximum width ~ 3 km. The intrusion, which was emplaced in folded carbonate and silicate metasedimentary rocks, consists of a medium-to coarse-grained

metagabbro composed mainly of dark-purple plagioclase, green pyroxene and green hornblende. However, composition, texture and structure vary greatly within the mass. Parts of it are fresh and unaltered or even slightly affected by metamorphism while others are highly altered, deformed and metasomatized. In less altered zones, the original igneous textures and structures are well preserved; in more altered ones, they are partly to wholly destroyed by the imposition of a strong cataclastic fabric.

The main mass consists generally of normal gabbro without olivine but some parts contain up to 23%. Typically, the slightly altered gabbro (Fig. 3, e.g., site 3, sample 5) consists of the following primary minerals: labradorite (44%), augite (48%), magnetite and pyrite (2%) and biotite (1%); the secondary minerals are: green hornblende (5%), calcite (tr) and sphene (tr). Biotite is molded on augite or iron oxide and is mantled, in turn, by green hornblende. The highly altered gabbro (Fig. 3, e.g., site 1, sample 2) consists of labradorite (28%), augite (5%), magnetite and pyrite (5%) and biotite (2%); the secondary minerals are: green hornblende (40%), scapolite (15%), calcite (4%), sphene (1%) and apatite (tr). Other minerals occasionally present in small amounts are: chlorite, epidote, spinel and tourmaline. In the altered gabbro, scapolite has partly replaced plagioclase (as grains or pods) and calcic pyroxene was altered locally to green hornblende (as grains or rims). Olivine has been found in 2 samples (Fig. 3, e.g., site 7, sample 2), where labradorite is present, reaction rims occurred between the 2 minerals to produce a fibrous corona texture. This texture exhibits 2 concentric shells around olivine. The inner rim consists of a colorless orthopyroxene (hypersthene?) and the outer rim of a pale green fibrous amphibole (actinolite?).

The Cheneaux metagabbro was emplaced evidently prior to metamorphism as an igneous body for it was affected obviously by deformation and metamorphism. It was involved in at least 1 episode of dynamic and thermal metamorphism. Further evidence for an igneous origin is found on its eastern boundary (near Portage-du-Fort) where it cuts across marbles and gneiss of the metasedimentary groups (Fig. 3, site 1). Here, the metasedimentary rocks display a narrow, well-defined contact metamorphic aureole, indicating emplacement into cool metamorphosed host rocks. Therefore a prolonged cooling history of tens of millions of years is unlikely.

Narrow dykes cut across the gabbro body. These dikes, ranging in thickness up to 20 cm, consist of a fine-grained metadolerite. Moreover, 2 dykes of a medium-grained metaalaskite have also been observed.

PALEOMAGNETISM

Sampling procedure

Some 53 oriented samples (188 specimens) were collected on 9 different sites (Fig. 3); the orientation was taken with a Brunton compass. The num-

ber of samples per site varied from 5 to 9 and the number of specimens per sample between 3 and 5. The majority of the samples were collected on the Cheneaux (CH) metagabbro. Four samples were collected in the country rock (Portage-du-Fort marble of Helikian age) for a baked contact test, 2 samples in dykes related presumably to the late phase of the gabbro intrusion and 3 additional samples in dolerite dykes which are younger than the gabbro. The oriented hand samples were cored in the laboratory. The sample locations were chosen by reference to a detailed geological map by Brun (1983, in press) at a scale of 1:5000 (Fig. 3). Efforts were made to select sites where there was an adequate exposure and where the structural geology was simple; great care was taken to avoid altered material.

Equipment used

The direction and intensity of remanent magnetization (NRM) were measured with a digital spinner magnetometer (model DSM-1) manufactured by Schonstedt Inst. Co. (sensitivity 10^{-4}-Am^{-1}). Alternating field demagnetization (AF) was carried out using a demagnetizer built at Université Laval having a maximum peak-to-peak intensity of 90 mT, the performance of which was improved largely by adding 3 large concentric mu-metal cylinders around the solenoid (Seguin, 1975). Thermal treatment of the specimens was done in a field-free demagnetizer (model TSD-1) manufactured by Schonstedt Inst. Co. The amount of residual ambient field present is 5 and 10 nT in the AF and thermal demagnetizer, respectively.

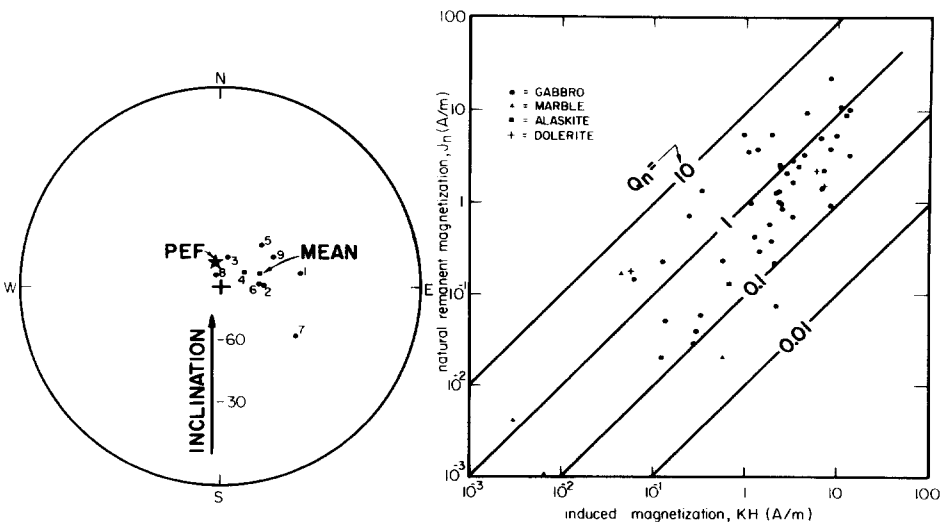


Fig. 4. Site mean NRM directions plotted on Wulff net for the CH metagabbro (left). Solid circles represent projections on lower hemisphere. The star represents the present Earth's field (PEF) at the sampling region and the square the mean NRM direction for all sites. Plots of sample mean intensity of NRM (J_n in A/m) against the induced magnetization (KH in A/m) with resulting slope lines indicating the values of Koenigsberger ratio, Q_n (right).

Remanent orientations and intensities

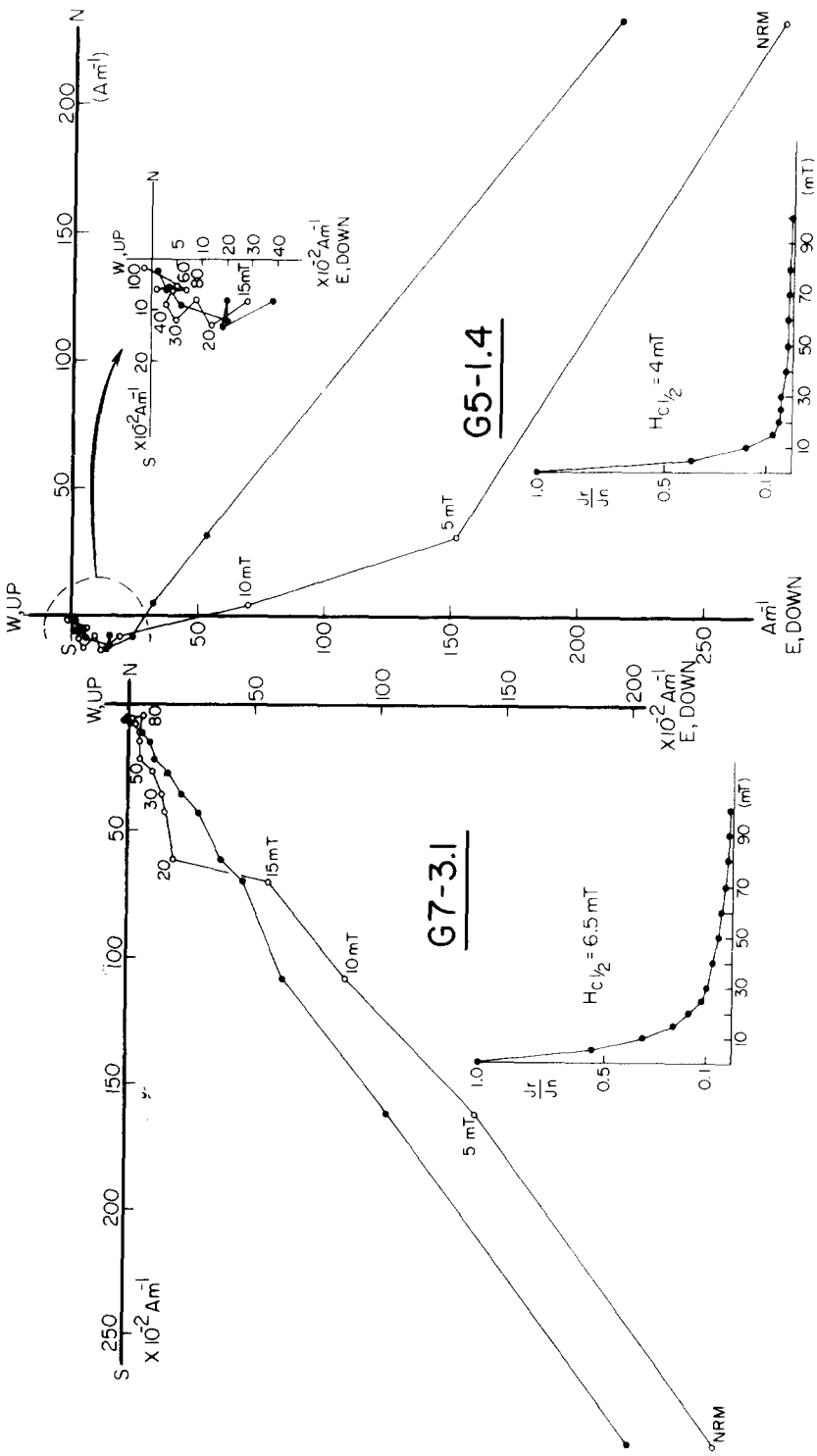
Site-mean NRM direction (D_0 = 72°, I_0 = 66°, N = 9, R = 8.47, α_{95} = 13.7°, K = 15) of Cheneaux (CH) metagabbro (Fig. 4), although characterized by wide scatter of positively inclined steep remanences, have a pronounced deviation away from the present Earth's field (PEF) (D = 347°, I = 75°) and a significantly different direction at the site localities, indicating that the CH metagabbro does carry some ancient remanence especially since the directions of the remanence of the few samples that lie close to the PEF were found to move away from those initial directions during subsequent AF or thermal demagnetization. The scatter in NRM with occasional dual polarity of remanences is therefore indicative of the presence of both low and high coercivity components. The intensities of NRM (Jn) vary from 10^{-3} to 10 Am^{-1} with values of Koenigsberger ratio (Q_n = Jn/KH) between 0.1 and 6 but with ratios predominantly between 0.3 and 0.9.

Preliminary demagnetizing studies

AF demagnetization

For the preliminary AF study, at least 4 specimens per Cheneaux site were demagnetized in 15 steps up to 90 mT, and vector diagrams (Zijderveld, 1967) were used in the subsequent analysis. For 95% of the samples, the intensities of magnetization at or before the 90 mT treatment had fallen to < 10% of the NRM intensity. In almost 50% of the samples the measurements indicated acquisition of spurious magnetization at higher demagnetizing fields (50—90 mT). Remanence behavior during AF treatment typical of most samples from the 9 Cheneaux (CH) metagabbro sites is shown by 2 specimens from two different sites (Fig. 5). Vector trajectories are linear until stable end-points are reached at higher fields. The linear trajectory for specimen G7—3.1 (site 7) indicates the presence of a stable, single component of magnetization with a southeasterly declination and shallow downward inclination as indicated by the linear decay of the intensity vector from 5 mT to the origin. The SE remanence is found to be a relatively hard component at sites 2, 3, 4 and 6 in the range: 30—80 mT with mean at 50 mT. For sites 1, 5, 7, 8 and 9 the best stability index is located in the range: 15—80 mT with a mean at 30 mT. Sites 2 and 7 contain only the SE component after AF demagnetization. An occasional soft component of magnetization directed towards the PEF (340°, intermediate to steep inclination) present in a few specimens (5—10) is removed by treatment of 5—10 mT.

The orthogonal plot for specimen G5—1.4 (site 5, Fig. 5) shows the presence of 2 components. One has a northeasterly directed magnetization having an intermediate downward inclination (D = 40°, I = 40°) which is removed apparently between 10 and 30 mT (more generally 15 mT). This component is predominant at sites 1 and 9 and is denoted NE remanence.


G5-1.4
G7-3.1
W, UP
E, DOWN
N
S
NRM
5 mT
10 mT
15 mT
$H_{c1/2}$ = 4 mT
$H_{c1/2}$ = 6.5 mT
$\frac{J_r}{J_n}$
(mT)
$\times 10^{-2}$ Am^{-1}
(A m^{-1})

Two less common components (W+ and W−) were isolated with AF demagnetization. The W+ remanence is present at sites 1 and 9; it is characterized by a WSW declination, an intermediate downward inclination and a coercivity in the 30—80 mT range (mean: 45 mT). This component is isolated after removal of either the SE or NE remanence but more frequently of the NE component. The W+ component may be either primary or secondary; with AF demagnetization only, it is impossible to decide one way or another. The W− remanence direction (WSW, intermediate upwards) was identified clearly after AF demagnetization on site 8 only. The W− remanence is almost antiparallel to the NE component and the coercivity ranges in the 20—60 mT (mean: 40 mT). This observation suggests that the W− remanence may be a secondary component as well. Thermal treatment may be useful in deciphering the complex nature of this component.

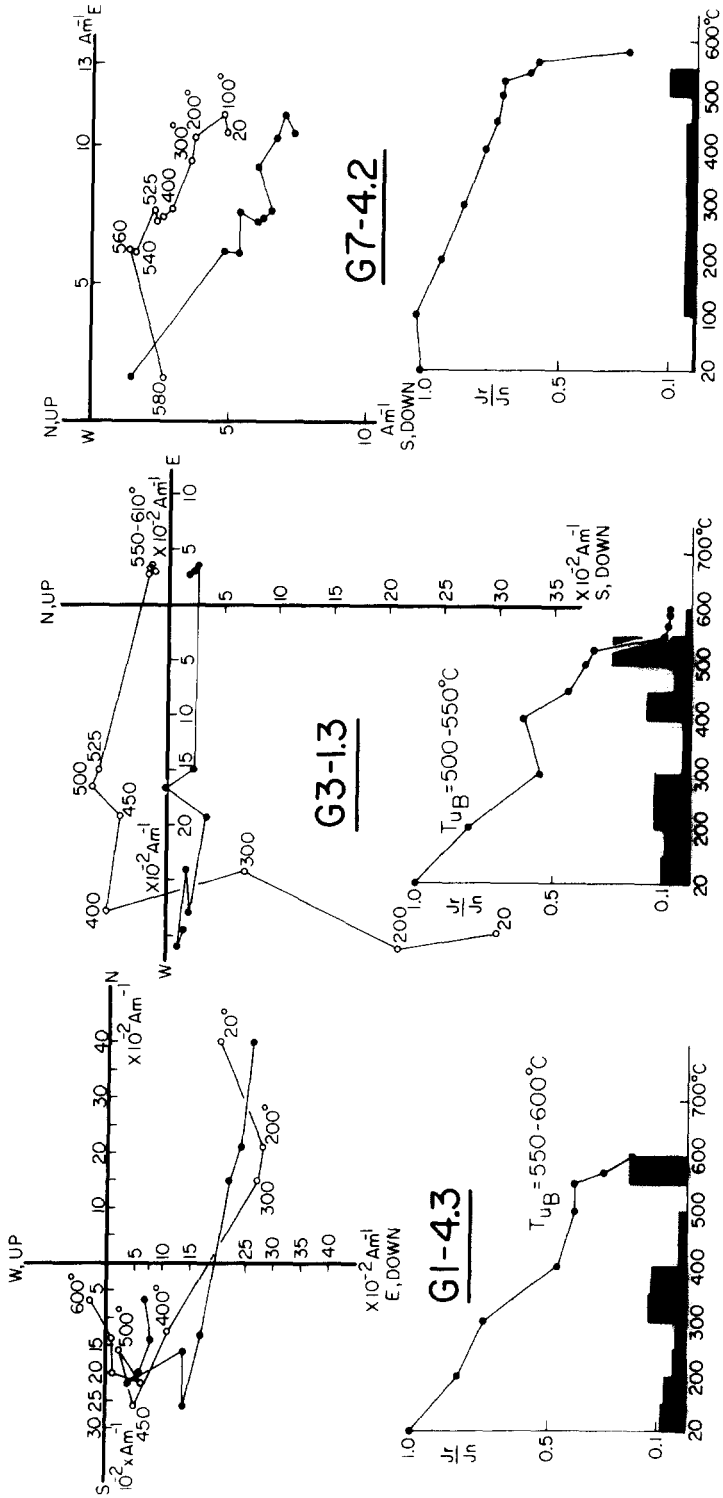
Thermal treatment

At least 14 specimens per site were demagnetized in 11 steps up to a temperature of 610°C. Preliminary thermal treatments on the CH gabbros revealed that the largest fraction of the specimens ($\lessapprox$ 75%) had only 1 component (SE) while the others had 2 and rarely 3. Figure 6 shows typical behaviors of 3 samples during thermal treatment from 3 different sites. In the case of specimen G7—4.2 (site 7), the vector trajectories are linear until stable end-points are reached at high temperatures (around 570°C). This behavior indicates the presence of a single stable component of magnetization with a southeasterly declination and shallow downward inclination. The SE component is about the same as the one isolated with the AF demagnetization technique (specimen G-7—3.1 (site 7) of Fig. 5). This SE remanence is the only one found on sites 2 and 7 but it is present on all other sites. Its blocking temperature is located in the 450—610°C range (mean: 560°C). The most common blocking temperature (T_B) is 570°C.

The orthogonal plot for specimen G1—4.3 (Fig. 6) shows the presence of 2 components. One has a northeasterly directed magnetization with an intermediate downward inclination (D = 25°, I = 45°) which is removed between 300 and 400°C in this case but more generally between 400 and 550°C (mean: 525°); the prevalent SE component is then isolated. The NE component is predominant at sites 1, 3, 6 and 9.

The W+ remanence is found on sites 4, 5, 6 and 8; its declination is WSW and its inclination shallow to intermediate. The component is erased generally in the 300—525°C temperature range; occasionally, it persists

Fig. 5. Representative AF demagnetization diagrams (Zijderveld, 1967) of successive magnetization vector for two CH metagabbro specimens as indicated. Solid (open) circles represent projections on horizontal (vertical) planes. Numbers next to open circles represent the demagnetization steps (in mT). Units indicated for axes are in A/m (10^{-3} emu cm^{-3}).


G7-4.2
N,UP
W
E
S,DOWN
Jr/Jn
G3-1.3
N,UP
W
E
S, DOWN
T_{uB}=500-550°C
G1-4.3
W,UP
E,DOWN
N
S
T_{uB}=550-600°C

at 550° and 570°C but with a polarity reversal. Most commonly, after disappearance of the W+ remanence, either the SE or W− components are isolated. An exception to this rule is depicted by the Zijderveld diagram of specimen G3—1.3 (Fig. 6). This exceptional specimen has recorded 3 components: (1) a W+ remanence in the 200—400°C range (2) a W− direction in the 400—525°C range and finally (3) a SE magnetization in the 550—610°C range. This specimen and a few other ones which behave in the same manner allowed the reconstruction of the magnetic chronology.

Finally, the W− component is encountered mainly at sites 3, 4 and 8. This component is usually isolated in the 400—550°C range. It may either persist to a higher temperature (600°C) as is the case for some samples at site 4 or decay allowing for the extraction of the SE component.

A total or 129 pilot specimens out of 188 were used for AF and thermal treatments. It was realized then that the thermal treatment was preferable to AF demagnetization. Consequently, the remaining 60 specimens were thermally demagnetized in 5 steps in the 500—600°C temperature range.

Statistical analysis of site-mean directions

Table I shows CH metagabbro site-mean directions for both AF and thermally cleaned results. Table II gives the SE, NE, W+, W− and NW remanence directions for the corresponding group-mean directions based on sites. The basis for determining these various components is: (1) similarities in NRM vector orientation from one site to the next (2) frequency, i.e., number of specimens belonging to a component (3) number of reversals (4) baked contact test when available (5) mutual relation of the gabbro and dyke components (6) blocking temperature spectra (7) median destructive field. The SE remanence is present in all gabbro sites as well as alaskites and marbles but in 1 out of 3 dolerite dykes only. The NE component is common to 7 of the 9 gabbro sites, the marble site and 1 of the 2 alaskite dykes. This component is not recorded in the dolerites. The W+ magnetization direction is found in 6 of the gabbro sites, 1 alaskite, 1 dolerite dyke, and the marbles. The W− component is present in 4 gabbro sites, 2 dolerite dykes and the marbles but it is absent in the alaskite dykes. The number of specimens (6) which recorded the NW direction is insignificant and it is concluded that this component is not a real one. Finally, 2 of the dolerite dykes recorded a different component, the direction of which is SSW with a shallow to intermediate inclination (Table II). This component is probably real as it is recorded in both dykes.

Fig. 6. Representative thermal treatment diagrams (Zijderveld, 1967) of successive magnetization vectors for 3 CH metagabbro specimens as indicated. The other symbols are as in Fig. 5. The blocking temperatures (T_B in °C) are indicated below the orthogonal plots.

TABLE I

Statistical summary of site-mean remanence directions at G, D, A and M sites

	SE Remanent component					NE Remanent component					W+ Remanent component				
	N(n)	D°	I°	K	α_{95}°	N(n)	D°	I°	K	α_{95}°	N(n)	D°	I°	K	α_{95}°
Site	Metagabbros (G)														
G1	5(13)	146	21	29.1	14.4	2(4)	44	38	16.2	—	2(3)	248	52	4	—
G2	5(12)	127	28	14.5	20.8	1(1)	28	57	—	—	1(1)	71	−13*	—	—
G3	4(12)	150	28	25.6	18.5	1(3)	80	28	—	—	—	—	—	—	—
G4	4(12)	120	34	23.1	19.5	1(1)	79	54	—	—	1(1)	235	10	—	—
G5	5(13)	134	25	11.6	23.5	1(1)	47	35	—	—	2(2)	249	22**	7	—
G6	5(13)	120	29	14.6	20.7	1(2)	49	48	—	—	1(2)	259	64	—	—
G7	5(14)	137	25	5.4	37.9	—	—	—	—	—	—	—	—	—	—
G8	4(9)	155	34	8.6	33.3	—	—	—	—	—	2(6)	289	73**	6	—
G9	3(11)	124	43	7.8	39.7	2(4)	24	42	82.4	27.7	—	—	—	—	—
	Metadolerites (D)														
D4	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
D8	1(4)	146	−23	—	—	—	—	—	—	—	1(1)	68	−20*	—	—
	Metaalaskites (A)														
A2	2(6)	150	19	12.7	—	1(1)	56	05	—	—	1(2)	68	−21*	—	—
	Marbles (M)														
M1	3(5)	149	19	37.4	20.4	4(7)	75	26	—	—	2(3)	279	36	2	—

N = Number of samples.
n = Number of specimens.
D° = Declination.
I° = Inclination.
K = Fisher's (1953) precision parameter.

TABLE II

Statistical summary of group-mean directions of magnetization and of paleomagnetic poles

Group	Metagabbros (G)						Metadolerites (D)				
	N(n)	%	D°	I°	K	α_{95}°	N(n)	D°	I°	K	α_{95}°
SE	9(109)	72	135	30	38.9	8.4	1	146	23	—	—
NE	7(16)	11	37	41	12.7	17.7	—	—	—	—	—
W+	5(15)	10	249	45	17.9	19.3	1	248	20*	—	—
W−	4(11)	7	258	−34	25.0	18.7	2	282	−43	—	—
SSW	—	—	—	—	—	—	—	216	34	134	21.7

N = Number of sampling sites.
n = Number of specimens.
% = Percentage of the specimens in this component.
D° = Declination.
I° = Inclination.

Chronological sequence of the components of magnetization

The SE remanence direction for the CH metagabbro could be older because it has the higher coercivity spectra, the higher T_B values and also because many specimens depict the following sequence of components upon increasing AF demagnetizing strengths and temperatures: (1) W+ (2) W− (3) NE and (4) SE. A possibly positive baked contact test is obtained on the in-situ country rock (marble) on site 1. In both the gabbro

W− Remanent component					NW*** Remanent component					SSW Remanent component				
N(n)	D°	I°	K	α_{95}°	N(n)	D°	I°	K	α_{95}°	N(n)	D°	I°	K	α_{95}°
1(2)	247	−38	—	—	—	—	—	—	—	—	—	—	—	—
—	—	—	—	—	1(1)	337	36	—	—	—	—	—	—	—
1(2)	278	−23	—	—	1(3)	340	10	—	—	—	—	—	—	—
1(3)	240	−34	—	—	—	—	—	—	—	—	—	—	—	—
—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
—	—	—	—	—	2(2)	350	18	19.3	—	—	—	—	—	—
—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
2(4)	266	−36**	272	15.2	—	—	—	—	—	—	—	—	—	—
—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
—	—	—	—	—	—	—	—	—	—	1(3)	214	30	—	—
2(3)	282	−43	—	—	1(1)	328	04	—	—	1(1)	219	39	—	—
—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
1(3)	292	−17**	—	—	—	—	—	—	—	—	—	—	—	—
3(6)	70	27*	53.2	17.1+	—	—	—	—	—	—	—	—	—	—

α_{95}° = Radius of cone of 95% confidence.
* = Reverse polarity.
** = Mixed polarity.
*** = Uncommon remanent component, group-mean directions not calculated in Table II.
\+ = Mean for lithologies: G, D and M.

Metaalaskites (A)					Marbles (M)					Paleopoles (G)+			
N(n)	D°	I°	K	α_{95}°	N(n)	D°	I°	K	α_{95}°	Long°	Lat°	dp°	dm°
1	158	19	—	—	1	149	19	—	—	32W	16S	5.2	9.3
—	—	—	—	—	—	—	—	—	—	37E	53N	13.1	21.5
1	248	21*	—	—	1	279	36	—	—	134W	6N	23.4	37.1
—	—	—	—	—	1	272	−43**	—	—	8E	21N	12.2	21.4
—	—	—	—	—	—	—	—	—	—	112W	18S	14.2	24.8

K = Fisher's (1953) precision parameter.
α_{95}° = Radius of cone of 95% confidence.
dp° = Semi-axe of ovals of 95% confidence.
dm° = Semi-axe of ovals of 95% confidence.
\+ = From the gabbros.
* = Reverse polarity.
** = Mixed polarity.

and the marble, the SE, NE, W+ and W− components are present; since both gabbro and marble recorded the same components, it is difficult to state that a pre-intrusion component was present in the marble. Consequently, this indicates that both formations recorded the same geological events during and/or after the gabbro intrusion. For small intrusive bodies of this kind, the contact metamorphic ring is generally small (2—3 m-thick). The NE and SE components are always present in the marbles even at large distances (> 5 m) from the intrusion contact. Components SW+ and

SW- are absent in the marbles at a distance > 3.5 m. This suggests that components NE and SE are regional overprints while components W+ and W- are related to events specific to the gabbro intrusive. This conflicts with the sequence of acquisition expected previously. The alaskite dykes and 1 dolerite dyke (D—8.5) recorded the same events as the gabbro with components: NE, SE and W+. This suggests strongly that the alaskite dykes and the older dolerite dykes are more or less synchronous with the metagabbro. The 2 other dolerite dykes (D—4.6 and D—8.4) did record a totally different geological event with a SSW downward shallow to intermediate remanence direction. These dykes are most likely younger. Murthy (1971), Palmer et al. (1977), Dankers and Lapointe (1981) and Lafleur and Hogarth (1981) have recorded younger events associated with these dykes.

Origin of magnetization

The NE and SE magnetizations, which are characterized generally by a high coercitive force (h.c.f.) and a high blocking temperature, contain both normal (N) and reverse (R) polarities. The R polarities (SE, shallow positive) are always present in the h.c.f. region of AF demagnetization and this implies that the reverse SE component has its memory mainly in fine-grained magnetite. There are a few (2 or 3) exceptions to this rule and then the normal (NW, shallow negative) component is isolated in the h.c.f. region (see for instance specimen G8—1.3 in Table III). The reverse SE magnetiza-

TABLE III

Typical examples of direction and intensity of residual magnetization after treatment

Specimen no: G2—2.2				Specimen no: G8—1.3			
(°C)	D (°)	I (°)	Jr (Am^{-1})	(mT)	D (°)	I (°)	Jr (Am^{-1})
20	085	53	2.32	0	314	65	2.61
100	082	53	1.96	5	307	57	1.36
200	125	55	1.58	10	304	55	1.29
300	095	52	1.54	15	284	−7	0.90
400	132	51	1.67	20	287	−32	0.68
450	101	45	1.16	25	287	−37	0.57
500	097	33	0.91	30	288	−45	0.53
550	098	34	0.85	40	286	−46	0.38
600	336	−34	1.17	50	285	−35	0.27
640	327	−40	1.01	60	289	−37	0.18
660	321	−37	0.97	70	292	−39	0.14
				80	329	−43	0.12
Dm = 328, Im = −37				90	311	−20	0.11
				100	333	−10	0.07
				Dm = 324, Im = −24			

tion isolated by thermal cleaning is largely predominant over the normal counterpart. SE magnetizations are extracted generally in the 450—550°C range whereas the NW normal component, though not as common, is present in the 560—540°C range (see for instance specimen G2—2.2 in Table III). The memory of the normal NW component is located at least partly in hematite. In summary, the SE component is characterized by mixed polarities but the reverse orientations (SE) are predominant over the normal ones (NW). A study of the opaque minerals using the optical reflecting microscope and microprobe analysis has shown that fine-grained pure magnetite is predominant. Ilmenite and titanomagnetite are observed in smaller quantities and hematite is observed rarely. The occurrence of the high T_B NW component still remains difficult to explain.

Significance of W+ and W− components

Thin sections were made on at least 2 samples per site (total: 38). A quantitative estimate of the plagioclase, olivine, clinopyroxene, orthopyroxene, amphibole, opaque minerals, accessory minerals (sphene, apatite, spinel, quartz and alteration products (scapolite, white mica, biotite, green hornblende, calcite, chlorite, epidote, tourmaline, etc. . .)) was done. The next step consisted in estimating the degree of alteration, the amount of deformation, the intensity of metasomatism and the amount of igneous textures destroyed. After examination of the thin sections, it is obvious that the rocks are heterogeneous when considering samples from the same site and sometimes within the same sample. This remark applies to sites 1, 2, 3, 4, 5 and 6 more specifically. In general, the degree of alteration varies from average to high, the amount of deformation from moderate to large (cataclastic texture), and the intensity of metasomatism from moderate to severe. The igneous texture is destroyed mainly in samples of sites 1, 3, 4, 5 and 6 but relatively well preserved in samples of sites 2, 7, 8 and 9. Site 7 is exceptional: it is characterized by a negligible degree of alteration, absence of deformation, very low intensity of metasomatism and a well preserved igneous texture; only the SE component is isolated on this site. Site 9 is characterized by a low to moderate intensity of metasomatism and amount of deformation; the original igneous texture is destroyed slightly but the degree of alteration is moderate to considerable. This feature may explain the occurrence of the NE component in addition to the SE component. Sites 2 and 8 depict a minor to medium degree of alteration (thus negligible NE component), a minor amount of deformation, a moderate degree of metasomatism and fairly well preserved original igneous texture. As site 8 contains both W+ and W− components of magnetization, it is logical to presume that they are related to regional metamorphism. When classified in order of increasing amount of deformation and intensity of metasomatism, the sites follow this sequence: 6, 5, 3, 4 and 1. In sites 6 and 5, the only additional component of magnetization to the SE and

NE directions is the W+ component. In site 3, the W+ component is absent and replaced by the W− component. On sites 1 and 4, both W+ and W− components were isolated. It is noteworthy that sites 1, 2, 4, 5 and 6 which are highly deformed (cataclastic texture) all contain the W+ component. This suggests 2 consecutive metamorphic events; an early one (W+ component) being related mainly to mechanical deformation (dislocation) and a late event (W− component) related probably to localized chemical changes. In any event, it is now certain that the W+ and W− components of magnetization are not associated with the primary intrusive event (TRM) but to secondary metamorphic events (PTRM, CRM, PCRM).

Paleopole positions and their relation to poles determined previously

The late Precambrian paleomagnetic field for North America has been investigated by various researchers and more particularly by Spall (1971), Helsley and Spall (1972), Irving et al. (1972), Elston and Scott (1973), Elston and Grommé (1974), Irving and McGlynn (1976), Morris and Roy (1977), Irving (1979) and Douglas (1980). Two main sources of paleomagnetic data were tapped for the interval −1200 to −950 Ma: (a) the Keweenawan rock series of the southern structural province of the United States and southwestern Ontario (e.g., Du Bois (1962), Books (1972), Henry et al. (1977), Roy and Robertson (1978), Watts (1981), Halls and Palmer (1981) and Palmer et al. (1981)) (b) the Grenville intrusives (mainly gabbros, ultrabasics and anorthosites) from southern Ontario for the most part, with a few studies from southcentral Quebec and Labrador (e.g, Hargraves (1959), Murthy (1971, 1978), Park and Irving (1972), Buchan and Dunlop (1973, 1976), Palmer and Carmichael (1973), McWilliams and Dunlop (1975, 1978), Ueno et al. (1975), Murthy and Rao (1976), Palmer et al. (1977), Buchan (1978), Symons (1978), Park and Roy (1979), and Palmer et al. (1979)).

A great many interpretations of the geometry of the APW of North America for this Precambrian period have been presented. The interpretation of paleomagnetic poles from the Grenville province represents one of the largest difficulties in this respect. Different models were put forward to explain the Grenville province. Some Grenville poles were interpreted as an integral part of the APW of Laurentia, others appeared discordant with the APW of Laurentia and involved large horizontal displacement and subsequent collision of a "Grenville plate". A summary of the ambiguities inherent to these 2 geological scenarios as well as a third one was presented by Palmer et al. (1979). Even though all the problems are far from being solved on this matter, it appears that for the time interval −1160 to −920 Ma, the Grenville poles are concordant generally with Helikian—Hadrynian poles obtained from rock units located outside the Grenville province. The plate tectonic implications are those of a pre-1200 Ma convergence of a possible separate continent "Grenvillia" with "Interior

Laurentia". It still remains to be proven that the separate "Grenvillia" plate existed before ~ −1150 Ma. Clearly, the argument about the Grenville belt having separated is not a logical consequence of our results.

As no radiometric dates are available for the CH gabbro, an attempt is made with the pole positions relative to the Grenville APW path to define the age of the gabbro (Fig. 7).

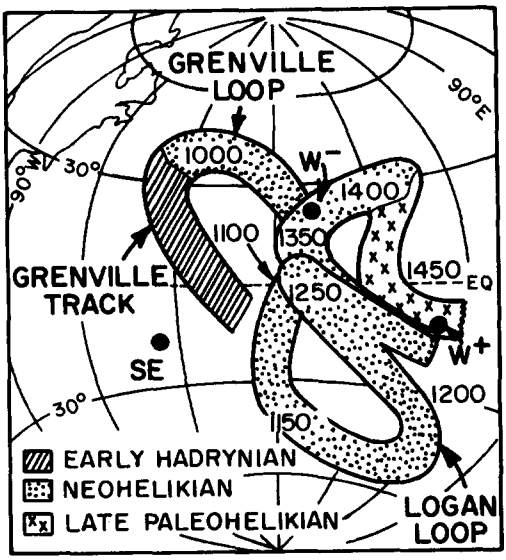


Fig. 7. VGP for the W+, W− and SE, remanence components plotted on Irving's (1979) APW curve for 1450—950 Ma interval. The VGP for the NE component is off the figure and is not shown.

The pole of component W− is 8°E, 21°N. Placed on the Neohelikian track (Irving, 1979), an age of −1075 to −1100 Ma is obtained. According to our interpretation, this terminal phase (timing of metasomatism) is preceded by a phase of mechanical deformation. The W+ component of magnetization represents either an early phase of mechanical deformation or the time of emplacement of the gabbro; its pole position is 46°E, 6°S.

It may be maintained that the argument that the rare W+ and W− magnetizations are the oldest is weak; however, this argument is based strictly on the concept of frequency. In a multi-orogenic belt like the Grenville province, multiple overprints are to be expected and, depending on the relative intensity of a metamorphic event, some components of magnetization are characterized by an apparent enhancement and others by an apparent rarity. We think that in such geological situations, the petrological and petrophysical parameters are more reliable indicators than the frequency of occurrence of a specific component.

Gabbro intrusives seem to occur within 2 time intervals in the Helikian era of the Grenville province. The first one spans −1300 to −1250 Ma while

the second spans −1180 to −1115 Ma. The age obtained from the paleopole position of the W+ component suggests that the Cheneaux metagabbro belongs to the first group. In spite of the fact that the SE component is present mainly in well preserved igneous textures, the baked contact test of this component is clearly negative and it is considered as an overprint. The pole (32°W, 16°S) of the SE component is located to the west of the descending Grenville track; its age of acquisition is estimated at ca. −975 Ma. The NE component is definitely a late event in the sequence of magnetization. Its antipole (37°E, 53°N; pole: 217°E, 53°S) is not really comparable to any well defined APW track. The best correlation is probably with the ill-defined late Hadrynian track (−750 to −800 Ma). Finally, the paleopole of the SSW component detected in the younger dykes is 258°E, 18°S. This pole position is similar to a late Hadrynian—Early Cambrian pole position obtained by Seguin and Lapointe (1983) in diabase and ultrabasic dykes intruding the Grenville rocks of the Montréal—Québec City region.

Thus, the age sequence of the magnetizations is based essentially on the various criteria established on the statistical analysis section, on the associated petrological and petrophysical characteristics and on a comparison with a standard polar wander path.

CONCLUSION

Even though commonly mechanically deformed and metasomatized, the CH metagabbro is not highly metamorphosed and for this reason some gabbro sites retained an old component of magnetization which is related apparently to the early deformation and is probably close to the primary (TRM) component which recorded the age of intrusion. The more deformed and metasomatized gabbro sites yield 2 components of magnetization (W+ and W−). The W− component, mainly related to metasomatism, is apparently younger; its estimated age on the Helikian—Hadrynian track (Fig. 7) is ~ −1075 Ma. According to the petrographic observations, the mechanical deformation took place before the metasomatic-event. The W+ component related mainly to this early deformation or the time of emplacement has an estimated age of ca. −1300 Ma on the Neohelikian track (Fig. 7). The direction of the SE component is found on 3 gabbroic bodies (Tudor, Thanet and Cheneaux). The corresponding isotopic age for the Tudor and Thanet B components are 1130 and ≒ 1050 Ma (Table IV). The difference between the paleopole obtained from this SE component and the established APW for the Helikian (1100—1300 Ma) may be due to: (1) inefficient and insufficient demagnetization (2) rotation of the gabbroic intrusive during the Grenvillian orogenic cycle. An age of −950 to −750 Ma is most probable for the acquisition of this SE component. As the SE component is predominant, an age span of ~ 300 Ma took place between the intrusive event and the paroxism of the Grenvillian orogenic cycle.

TABLE IV

Paleomagnetic poles discussed in the present study

Rock unit and location	Age (isotopic)	Component	polarity	Dec°	Inc°	Lat°	Long°	Pole	Reference
Ottawa basic intrusion			N+R	271	−66	32S	155E	Ot	Irving et al. (1972)
Frontenac dykes, Ont.	751∓74 and 817∓70 Ma		R	103	50	12S	162E	Fr	Park and Irving (1972)
Wilberforce pyroxenite, Ont.			N	293	−60	14.5S	148E	Wi	Palmer and Carmichael (1973)
Tudor metagabbro, Ont.			N	324	−37	17.1N	137E	Tu	
Grenville Front anorthosite, Ont.			R	117	25	8.2N	161.3E	Gf	
Haliburton intrusions, Ont.		A	N	266	−73	39S	144E	Hb	Buchan and Dunlop (1973)
		B	N	299	01	21N	171E	Hb	
		C	R			1.5S	165E	Hb	Buchan (1973)
Magnetawan Iron metasediments, Ont.			N	302	−73	24S	130E	Mg	McWilliams and Dunlop (1975)
Whitestone intrusions, Ont.		W	N	285	−56	16S	156E	Ws_1	Ueno et al. (1975)
		Z	N	292	−67	22S	146E	Ws_1	
		Y	R	104	36	05S	168E	Ws_2	
		X	R	122	45	01S	152E	Ws_2	
Indian Head anorthosite, Newf.			N	311	−58.5	8.0S	157.5E	In	Murthy and Rao (1976)
Steel Mountain anorthosite, Newf.			R	163	32.5	22.5N	139E	St	
Haliburton intrusions, Ont.		A	N	272	−73	35.6S	142.5E	Hb_1	Buchan and Dunlop (1976)
		B	N	300	10	24.5N	172.3E	Hb_3	
		C	R	105	35	3.0S	167.4E	Hb_2	
Umfraville metagabbro, Ont.	1180∓20 Ma		N	100	45	11S	166E	Um_1	Symons (1978)
Thanet metagabbro, Ont.		A_1	R	91	62	28S	158.4E	Th_1	Buchan (1978)
		A_2	R	78	55	32.3S	172E	Th_2	
		A	R	84	59	30.4S	165.3E	Th_3	
		B	N	308	−13.3	20.3N	159.3E	Th_4	
Umfraville metagabbro, Ont.	1160 and 911 Ma	A	N	279	−58	21S	157E	Um_2	Palmer et al. (1979)
		B	N	110	42	5.0S	161E	Um_2	

The highly cataclased alaskite dykes (A) are characterized by 2 components of magnetization (SE and W+). This indicates that the alaskites are almost synchronous with the metagabbro intrusion and it suggests strongly that the mechanical deformation on the time of intrusion is responsible for the existence of the W+ component in the alaskites. The same conclusion applies to one of the dolerite dykes (in site 8). This is not so for other dykes at sites 4 or 8. The SE component obtained on these dykes is radically different from all other directions of magnetization. It is thus concluded that 2 sets of dyke of different age are present. The younger mafic dyke rocks are fine-grained amphibolites altered and somewhat metasomatized. The paleopole position corresponding to this SE component of magnetization is 112°W, 18°S (Table III). Few poles of the Hadrynian track fall in this region (Roy and Robertson, 1978; Irving, 1979). However, the antipole (68°E, 18°N) is located to the east of the descending part of the Hadrynian track in the −800 to −670 Ma age span. It appears that a larger excursion to the east is taking place near the equatorial zone of this segment of the Hadrynian track; a −700 to −800 Ma age span can be assigned to these late mafic dykes (Seguin and Lapointe, 1983).

Finally, this paleomagnetic study demonstrates the contribution of petrology and geochemistry to the understanding of successive geological and related magnetic events in Grenville rocks.

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